Biomedical & Mechanical Engineering Portfolio

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Stance Control Orthosis

Team Goal: develop a knee brace that automatically locks during the stance phase of gait and unlocks during swing phase. The device integrates a mechanical locking joint with sensors that detect gait phase. The goal is to improve mobility and safety for individuals with lower-limb weakness or impairment.

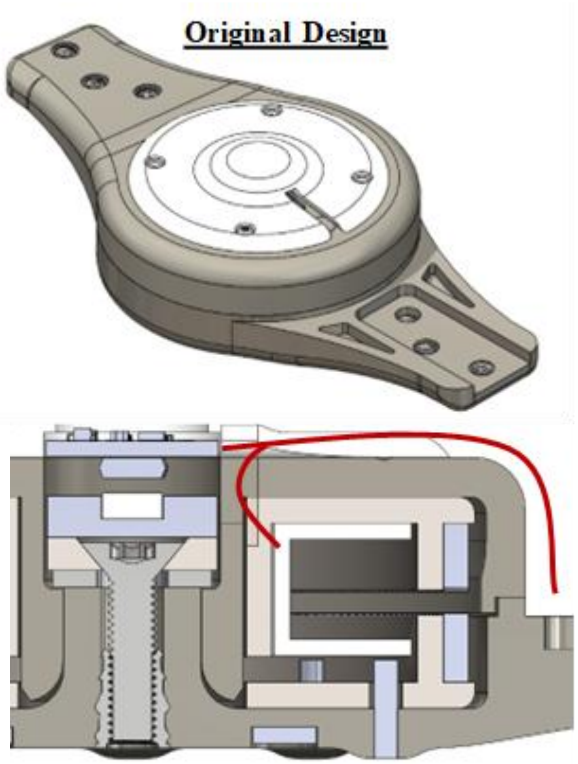
My Role: Mechanical Redesign

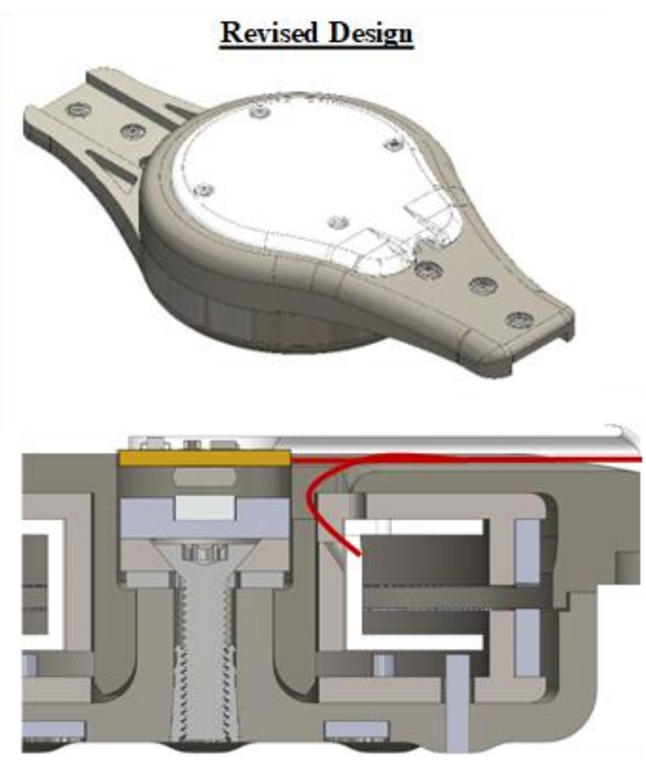
Problem: Wires controlling the locking mechanism repeatedly failed due to strain and sharp bending at the joint

Objective: Redesign the locking joint to improve wire routing and provide strain relief to prevent mechanical failure

Solution:

- Redesigned the locking joint in **SolidWorks**
- Implemented strain relief features to protect wires
- Evaluated wire routing paths to reduce bending stress
- Iterated design based on testing and feedback





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Stance
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Stance Control Orthosis

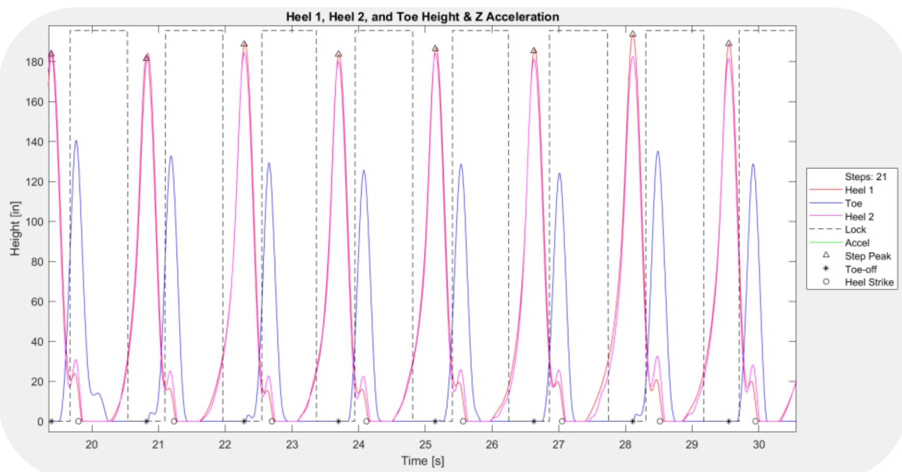
My Role: Clinical Testing and Data Processing

Objective: Develop clinical testing protocol and software script to evaluate the reliability and effectiveness of the SCO's locking mechanism.

Clinical Trial:

- Designed clinical trial protocol to evaluate the device
- Obtained **IRB approval**
- Administered the clinical trial to 10 able bodied subjects

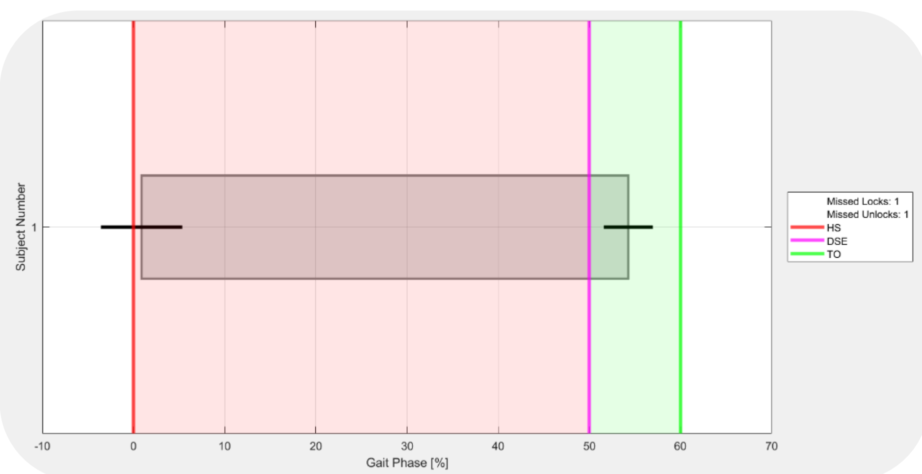
Example of testing and clinical trial set-up



Example of fully characterized gait data graph output

Data Processing:

- Developed **MATLAB** code to:
  - Synchronize motion capture and brace data
  - Fully characterize gait events
  - process synchronized locking and **motion capture lab** data
  - Generate and document final performance metrics through graphs



Example of statistical summary graph displaying testing and clinical trial results

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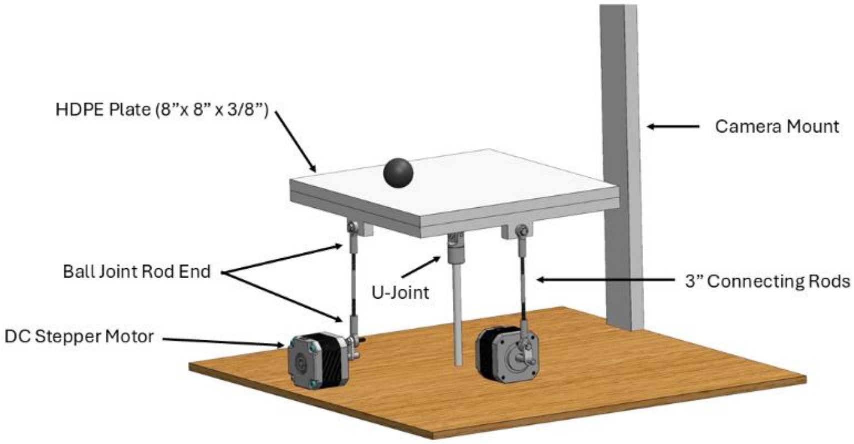
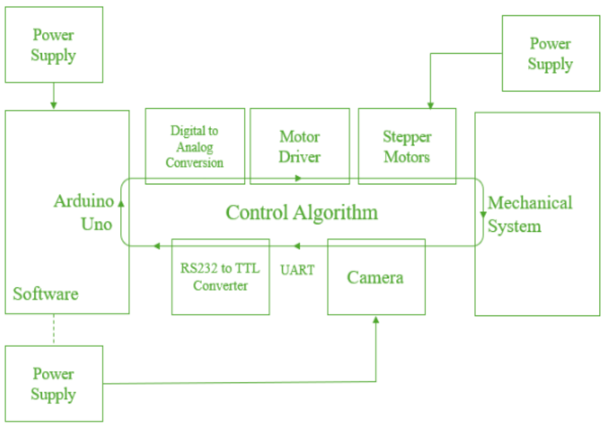
Ball Balancing Plate

Team Goal: Develop a device which controls the position of a ball on a plate using real-time feedback. An optical sensor tracked ball position, and motors adjusted plate orientation to control motion.



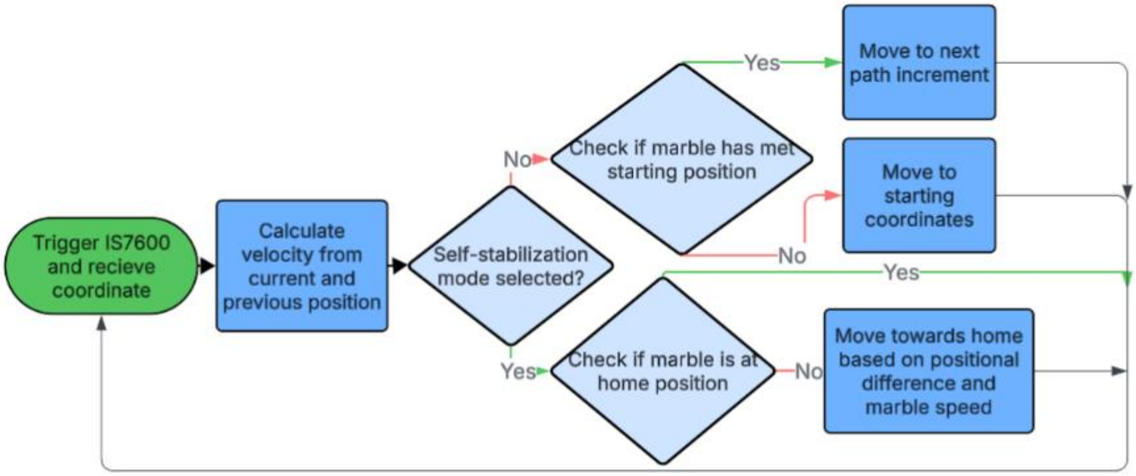
System Design



System Architecture

My Role: Software Development

- Developed **Arduino** code to receive position coordinates from optical sensor
- Calculated ball velocity from real-time data
- Implemented a **closed-loop control system** to adjust motor positions
- Integrated sensor input, control logic, and actuation
- Programmed two modes:
    - Stabilization: hold ball at center
    - Circular motion: move ball in a continuous path
- Tuned system for stability and responsiveness



Software Flow Chart

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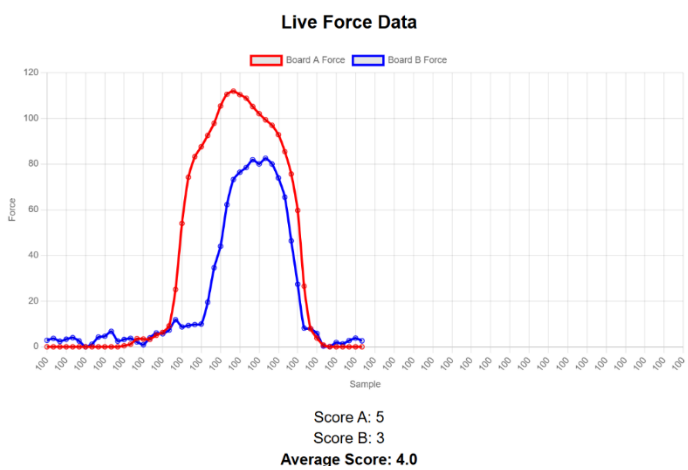
F.O.R.C.E Instrumented Crutch System

Team Goal: Design and test a prototype device capable of accurately measuring, processing, and reporting loading forces on assistive walking devices, enabling clinicians

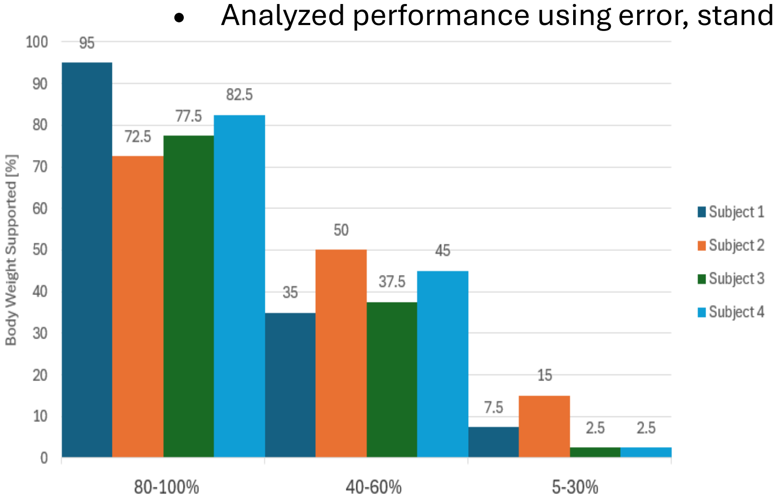


Real time graph of device implemented on two crutches

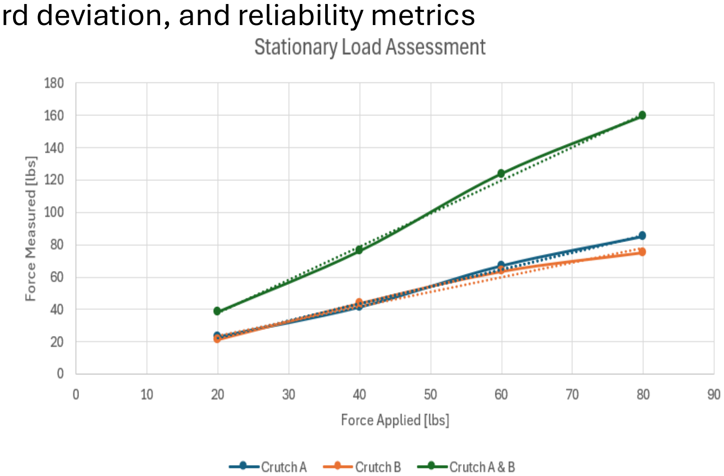
Device Design

My Role: Software Design and Testing

- Developed **Arduino** software to convert strain gauge data into force values
  - calibrated force readings using experimental data and **linear interpolation**
- Implemented **band-pass filtering** to reduce noise and signal drift
- Programmed algorithms to detect loading events and peak force per step
- Created a scoring system based on percentage of body weight applied
- Supported multi-device data streaming to an external device for real-time monitoring using **Python** and **Bluetooth**
- Designed and executed static and dynamic testing protocols
- Analyzed performance using error, standard deviation, and reliability metrics



Dynamic Load Assessment Results: Device reported weight within a specified percentage range precisely but slightly inaccurately



Stationary Load Assessment: Device reliably and accurately reported loading applied to the device

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VOITH

Voith Hydro: Ludington Project

Project Goal: Begin preparation to refurbish system of 8 turbines: design new components, design needed removal and reassembly tools, and detail disassembly, refurbishment, and reassembly procedures.

**** Due to confidentiality, no pictures of my work could be disclosed****

Lifting the Turbine Bottom Ring

Problem: Separate the turbine's bottom ring from the assembly considering the restrictions of limited space and safety concerns for personnel.

Objective: Develop safe, practical, and structurally sound methods for lifting and suspending the bottom ring using jacks and support structures.

Solution:

- Developed and proposed two different lifting methods
- Calculated quantity and size of jacks needed
- Designed jacking stands and strong back support plates in **Solid Edge**
- Performed structural and weld stress calculations
- Determined appropriate **safety factors** to include in system component design
- Conducted a **design review** to propose each method to the client

Lifting and Securing Turbine Housing

Problem: Lift and secure turbine housing to the guide bearing housing (directly above it) while avoiding structural failure or damage.

Objective: Design a safe method to lift and secure the turbine housing.

Solution:

- Designed an extendable jacking stand in **Solid Edge**
- Determined number and size of jacks required
- Designed two versions of securing clips
- Determined appropriate **safety factors** to include in system component design
- Conducted a **design review** to propose each method to the client